Arrays in Java

23 May 2026 08:59

WAPJ to store the marks of 50 students of class 10. Your program will find what the highest marks and what is the lowest marks, we need find number of student have got marks lower than class average.

Data type of marks is of type number(byte, short, int)

We have to use the concept of Array in Java.

What is an Array?
An Array in Java is a collection of elements of the same data types stored in contiguous memory locations. Arrays are used to store multiple values in a single variable instead of declaring separate variables.

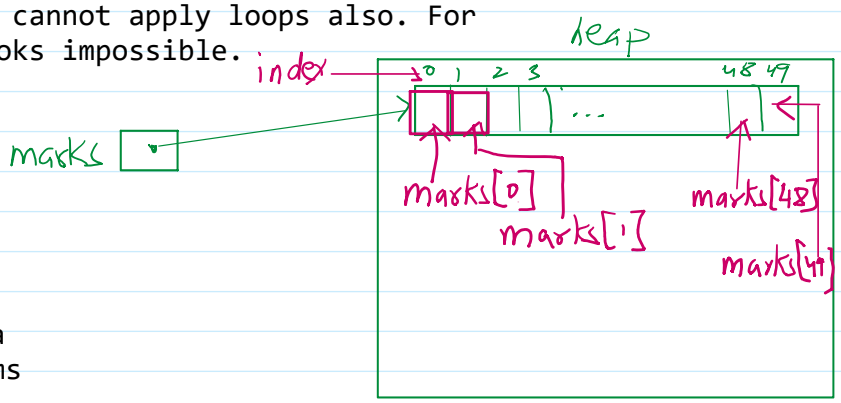
Why Arrays are needed?

If you see the above problem, it cannot be solved by declaring multiple variables. Therefore we need array to solve it.

int m1, m2, …, m50;// to feed data into these variables we need to write multiple repeated statements, even we cannot apply loops also. For comparing these variables data, it looks impossible.

With arrays:

```
int marks[] = new int[50];
```



Array make programs:

- Easier to manage
- Faster for processing large data
- Suitable for loops and algorithms

Features of Arrays

- Store multiple values of same type.
- Fixed size
- Index starting from 0.
- Can store primitive and object data types.
- Faster data access using index.

Syntax of Array Declaration

```
data_type[] arrayName;
        or
data_type arrayName[];
```

```
int[] marks;
    or
int marks[];
```

Array Creation

```
     marks = new int[50]; // This creates an array of size 50.

     Or
Data_type[] variable_name = new Data_type[size];
             Or
Data_type variable_name[] = new Data_type[size];

int[] marks = new int[50];
        or
int marks[] = new int[50];


Data_type[] variable_name;// it means in that row all the variables are
array type.

Data_type variable_name[]; // its means only that variable is array
type.
```